

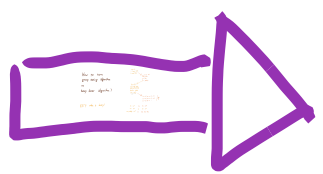
A Unified Approach for Group Testing Heavy Hitter Compressed Sensing Multiple Access Channel

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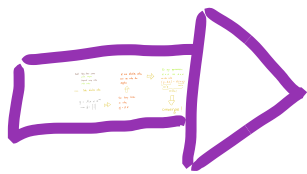
w/ Venkatesan Guruswami

Based on
[2301.08283]

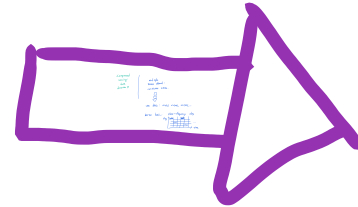
Group Testing



Heavy Hitter



Compressed Sensing



Multiple Access Channel

Background



IDEA



Mathematical Formulation

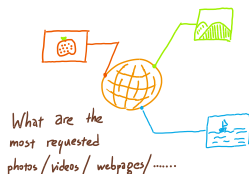
$\mathcal{X} \subseteq \{0,1\}^n$
 $\mathcal{Y} \subseteq \{0,1\}^m$ if $\mathcal{Y}$ is given is $\mathcal{X}$
 $A \in \{0,1\}^{m \times n}$
"testing matrix"
 $A_{ij} = 1$ if $\mathcal{Y}_i \cap \mathcal{X}_j \neq \emptyset$

Example

$A = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}$

What can A do?

What if there are $\gg 2^n$?



What are the most requested photos/videos/ websites/.....

Easy: ask a counter "no need to log and sort!"

view count

2,700,122

view count

3,111,574

Actually, not a good idea

Most resources are requested $\ll$ 10 times

We are missing a lot of times

counting / sorting 12345678910

IDEA

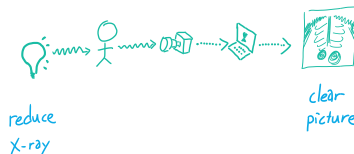
$x = \begin{bmatrix} 1000... \\ 1000... \\ 1 \end{bmatrix}$ view count vector

Fold x by A

$y = Ax$

to save space

But does it save time?



reduce X-ray

clear picture

Mathematical Model

$$\mathcal{Y} = A \in \mathbb{C}^{m \times n}$$

$$\mathcal{X} = x \in \mathbb{C}^{n \times 1}$$

$$\mathcal{Y} = y = Ax \in \mathbb{C}^{m \times 1}$$

solve x

Caveat: Sometimes x is too large
We cannot use linear algebra!
(if $n \gg m$)

$$\min_x \|y - Ax\|_2 \leq \lambda \|x\|_2$$

$$\min_x \|y - Ax\|_2 \geq \lambda \|x\|_2$$

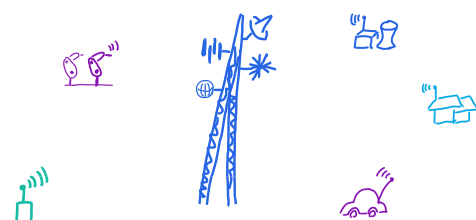
Traditionally, there will work if A is "isometry", i.e. $\|Ax\| = \|x\|$

Why? Because if $x \neq 0$ then $Ax \neq 0$ then $\|y - Ax\|$ is big

And then there are people trying to corrupt isometries

$$A = \begin{bmatrix} 1 & 1 & 1 & \dots \\ 1 & 1 & 1 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

$$A = \begin{bmatrix} \text{Gaussian} & \dots \\ \text{Gaussian} & \dots \\ \vdots & \ddots \end{bmatrix}$$



$$y := \sum_j e^{s_j + i p_j} \cdot M_j$$

s_j : strength of signal

p_j : phase / sync error

M_j : message

Caveat: even larger n is # possible message

Caveat: we don't know who is active/who is idle

Caveat: If two signals collide they destroy each other



Caveat: Dealing with continuous waves